

Dose-Aware MR-to-CT Synthesis through a Differentiable Physics-Prior Residual 3D U-Net for Photon Dose Prediction on 0.35 T MRI (DoseRAD2026 Photon-MRI Task)

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Abstract. We predict photon control-point dose directly from 0.35 T MRI by making synthetic-CT (sCT) generation *dose-aware*. A classifier-refiner synthesizer maps the MR to a density image; an analytical, differentiable physics operator turns density and control point into six dose-shaped channels; and a residual 3D U-Net corrects the analytical prior toward Monte Carlo. Because the physics is differentiable in the density, the synthesizer is trained *jointly through it* with the dose loss; the sCT is optimized for the dose, not for image fidelity. This matters: across five coarse-prior variants, standalone image quality did not predict fused dose accuracy ($r=0.02$), and a TotalSegmentator-MRI lung mask with far better Dice than our classifier gained nothing dosimetrically. The full chain lifts $\gamma 1\%/1\text{ mm}$ from 68.7 (bulk-density) through 82.6 (image-fidelity sCT) and 88.8 (dose-aware synthesis) to 90.7 (joint training) on the development fold; 5-fold cross-validation gives 91.1 ± 1.6 , and a cross-validated real-CT ceiling of 95.8 ± 1.3 isolates the remaining synthesis cost at 4.7 ± 1.4 points. On the hidden test set, MR intensity shift between training and test images emerged as the dominant failure mode and was substantially mitigated by retraining the tissue classifier with strong intensity-shift augmentation. The final full-data submission scored $\gamma = 84.11$, per-beam MAE 0.0172, IDD 0.0127, DVH 1.00 at 94.8 s per job; the fastest runtime on the preliminary leaderboard at submission time.

Keywords: MR-only planning · synthetic CT · dose-aware synthesis · differentiable physics · MR-linac

1 Introduction

On MR-linacs and in MR-only workflows the planning image is an MRI, whose intensities encode neither electron density nor attenuation. The established remedy (synthesize a CT, then compute dose) optimizes the synthesis for an *image* objective (MAE, SSIM, Dice) that is only a surrogate for the clinically relevant quantity [1,2,3,4,5,6]. We instead couple synthesis and dose in one differentiable

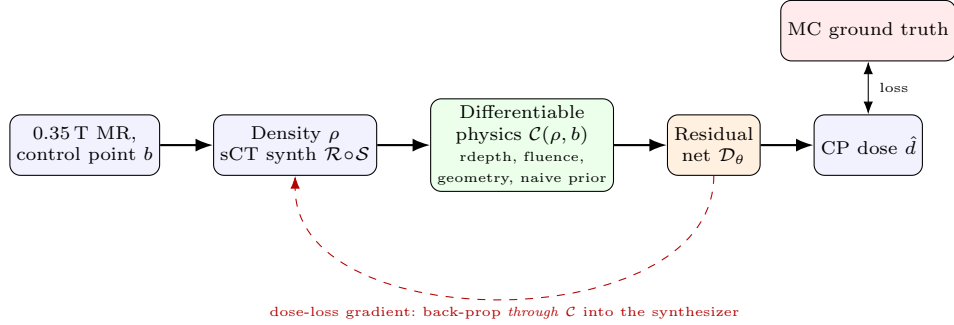


Fig. 1. Dose-aware MR-only pipeline: the synthesizer is trained jointly through the differentiable physics with the dose loss (dashed), so the sCT serves the dose, not an image metric.

pipeline. An analytical physics operator $\mathcal{C}$ maps density and control-point descriptor to dose-shaped channels; a residual 3D U-Net $\mathcal{D}_\theta$ corrects the analytical prior toward Monte Carlo (MC); and because $\mathcal{C}$ is differentiable in the density, a synthesizer $\mathcal{X} = \mathcal{R} \circ \mathcal{S}$ placed upstream is trained *through the physics* with the dose loss. The synthesizer is thereby free to place density where the *dose* needs it rather than where an image metric would reward it; and our central empirical finding is that these two objectives genuinely diverge (Sec. 3.3). This report details the Photon-MRI system; the same framework serves our submissions to all four DoseRAD2026 tasks.

2 Methods

2.1 Data

Official DoseRAD2026 training set [10] (75 patients: 39 thorax, 36 abdomen; SynthRAD2025 cohort [11]): per patient a co-registered 0.35 T MR (ViewRay MRIdian, bSSFP, breath-hold) and planning CT, beam parameters, and per-control-point MC dose-to-medium (matRad [9]) on a $2 \times 2 \times 2$ mm grid; 540 control points per patient. Per the challenge definition, the MRI-task ground truth is the CT-computed MC dose mapped onto the co-registered MR, so the achievable target for MR-only calculation is the CT dose itself; the framing behind our real-CT-ceiling analysis (Sec. 3.3). The sCT refiner additionally uses 166 field-matched 0.35 T MR–CT pairs from the public SynthRAD2025 pool (disjoint from validation patients); no other external data. Development: 5-fold CV over all 75; **the final submission is retrained on all 75**, with the leaderboard as held-out validation.

Photon-MRI (skin-entry, MR->sCT->dose) - 1ABB006 CP 0_134 (abdomen) | prior<->GT corr 0.899 pred rel-MAE(>5%) 2.8%

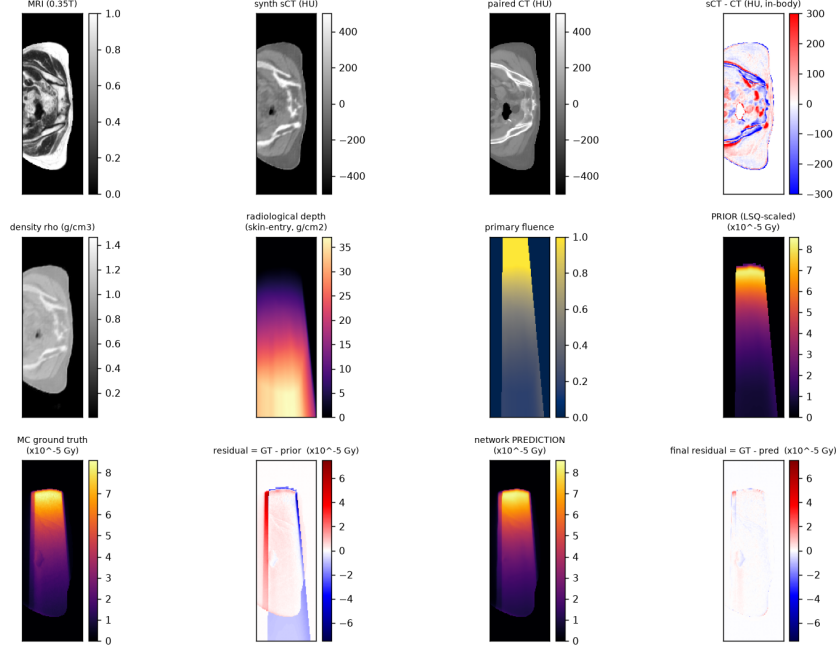


Fig. 2. Deployed Photon-MRI chain (abdomen, 1ABB006). Row 1: synthesis ; MR (0.35 T) $\rightarrow$ synthetic CT $\rightarrow$ paired real CT $\rightarrow$ in-body sCT-CT residual. Remaining panels: physics channels on the synthetic density $\rightarrow$ LSQ-scaled naive prior $\rightarrow$ MC ground truth $\rightarrow$ residual (GT-prior) $\rightarrow$ prediction $\rightarrow$ final residual (GT-pred).

2.2 Model

Synthesizer $\mathcal{K} = \mathcal{R} \circ \mathcal{S}$: $\mathcal{S}$, a whole-image 3D U-Net classifier, assigns each voxel to air/lung/soft-tissue/bone (CT-HU band edges $-500/-300/+200$) and maps classes to a bulk-HU coarse CT; $\mathcal{R}$, a whole-image refiner, maps [MR, coarse] to the sCT, from which the official HU calibration yields density. **Physics** $\mathcal{C}$: the six photon channels (density, ray-marched radiological depth, MLC-projected primary fluence, distance-to-CAX, source distance, naive analytical surrogate), all differentiable tensor operations, with the per-ray skin-entry rule (depth accumulates from the first $\rho > 0.05 \text{ g/cm}^3$ crossing; prior masked strictly upstream). **Dose network** $\mathcal{D}_\theta$: DoseUNet3D (base 48; four levels; ASPP bottleneck; Soft-plus head; FiLM modality conditioning), warm-started from the Photon-CT network. Figure 2 shows the full deployed chain: synthesis quality and its dosimetric consequence in one figure.

Photon-MRI (skin-entry, MR->sCT->dose) - 1THB048 CP 2_133 (lung) | prior->GT corr 0.837 pred rel-MAE(>5%) 6.2%

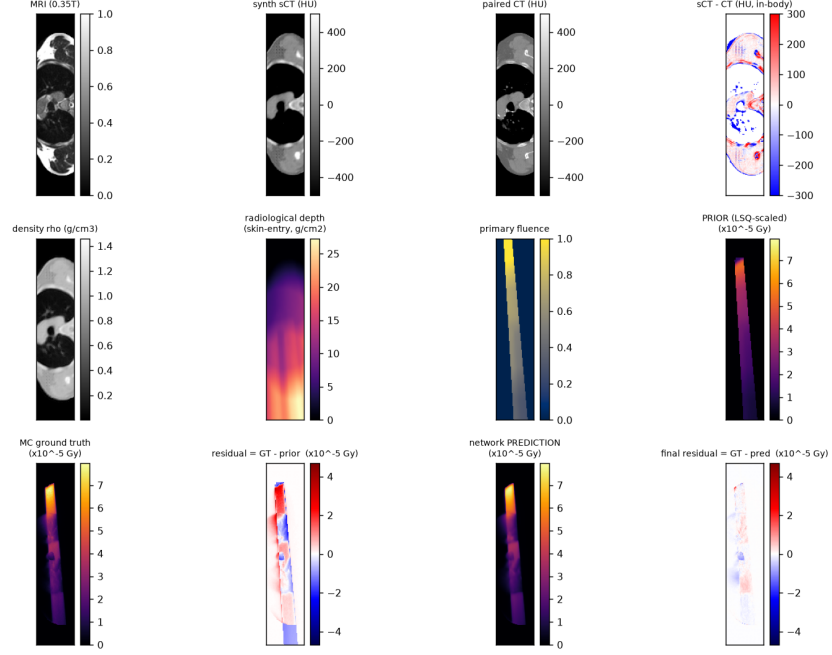


Fig. 3. As Fig. 2, lung case: 0.35 T MR carries little lung density information; lung is the site of the largest synthesis cost.

2.3 Training

Stage 1: the tissue classifier and refiner are pretrained (the refiner also on the 166 field-matched pairs); the dose network is warm-started from Photon-CT. Stage 2 (dose-aware): refiner and dose network train *jointly*, 15–20k steps, with the dose loss back-propagating through $\mathcal{C}$ into the refiner, plus a region-weighted per-voxel sCT anchor to the real CT as regularizer. Loss and optimization as in our CT system (high-dose-weighted L_1 , AdamW, EMA, patches 128^3). **Robustness for the hidden test:** the dominant held-out failure mode was MR *intensity shift* between our training MRs and the hidden-test MRs (diagnosed by leaderboard probes: internally synthesized shifts reproduced the drop, and the shared tissue classifier accounted for most of it). The deployed classifier is therefore retrained with strong MR intensity-shift augmentation (bias/gamma/scale perturbations); no augmentation is used in dose training itself. Reproducibility: fixed seeds, YAML configs, EMA checkpoints; code and weights to be open-sourced (Sec. 6).

Table 1. Photon-MRI results. CV: 5-fold over the 75 public patients. Hidden test: official metrics of the final full-data submission (August 2026). Gate: 181 s/job.

Metric	Value
CV plan $\gamma 1\%/1\text{ mm}$ (5-fold)	91.1 ± 1.6
CV real-CT ceiling / synthesis cost	95.8 ± 1.3 / 4.7 ± 1.4
Hidden-test per-beam masked MAE	0.0172 ± 0.0060
Hidden-test IDD curve distance	0.0127 ± 0.0079
Hidden-test stratified plan MAE	0.0129 ± 0.0070
Hidden-test plan $\gamma 1\%/1\text{ mm}$	84.11 ± 7.73
Hidden-test DVH clinical score	1.00 ± 0.77
Hidden-test runtime per job	94.8 s (fastest on the board at submission; sCT $\sim 2\text{ s/image}$)

2.4 Evaluation

Internal: plan-level local γ [8] (pymedphys, $1\%/1\text{ mm}$, $\geq 10\%$ Rx cutoff, byte-identical to the vendored official code), per-beam masked MAE, official IDD and stratified-MAE reimplementations; a frozen 16-patient held-out cohort gates every deployment change (*no accuracy loss, no speed loss*). Mechanistic probes: the *real-CT ceiling* (the trained MRI pipeline re-run with the real CT substituted for the sCT, isolating the synthesis cost patient-matched) and a five-variant coarse-prior sweep relating image fidelity to fused dose accuracy. Runtime: official $T = t_{\text{fix}} + N \cdot t_{\text{dose map}}$, gate 181 s, double-weighted.

3 Results

3.1 Accuracy: cross-validated and hidden-test

Per-fold CV $\gamma 1/1$: 90.7/90.2/92.2/88.9/93.6 (mean 91.1 ± 1.6). Development fold ($n = 16$): $\gamma 1/1 = 90.7 \pm 5.7$ (abdomen 94.6, lung 86.8), $\gamma 2/2 = 97.7$, $\gamma 3/3 = 99.1$; per-beam masked MAE $1.55 \pm 0.38\%$. Table 1 adds the hidden-test metrics. The CV-to-hidden-test gamma drop ($91.1 \rightarrow 84.1$) is dominated by MR intensity shift (Sec. 2.3): on the sister proton-MRI task the shift-augmented classifier recovered +3.1 board gamma, and the same fix is deployed here.

3.2 Ablation: each dose-aware stage pays

Table 2: bulk-density assignment through the engine reaches 68.7; a learned image-fidelity sCT 82.6; making the synthesis dose-aware (frozen engine) 88.8; joint training of synthesizer and dose network 90.7. The two learned-synthesis stages are complementary; the coarse prior lifts lung, dose-aware training lifts abdomen.

Table 2. Photon-MRI ablation (plan γ 1%/1 mm, development fold, $n = 16$).

configuration	ALL	abd	lung
bulk-density coarse CT $\rightarrow$ engine	68.7	80.4	56.9
learned sCT (image-fidelity) $\rightarrow$ engine	82.6	89.5	75.7
+ dose-aware synthesis (frozen engine)	88.8	93.7	84.0
+ joint dose-net training (deployed)	90.7	94.6	86.8

3.3 Image fidelity does not predict dose accuracy

Across five coarse-prior variants passed through the dose-aware fusion, standalone bulk- γ was uncorrelated with fused dose γ (Pearson $r=0.02$; Fig. 4). The extremes make the point concretely: the *worst* standalone variant (a soft-tissue-only bulk map, standalone γ 61.1) gave the second-best fusion result (90.4), while the variant with the *lowest* classifier Dice achieved the best fusion (90.7). A TotalSegmentator-MRI [7] lung mask (Dice 0.79 vs. our classifier’s 0.33) did not improve dose. The mechanism is exactly the joint training: gradients from the dose loss place density where the dose needs it, independently of coarse quality. This directly cautions against selecting sCT models by MAE/SSIM/Dice for dosimetric use. The cross-validated real-CT ceiling (95.8 ± 1.3) bounds what any synthesis can add: the remaining photon synthesis cost is 4.7 ± 1.4 points, roughly half the corresponding proton cost (8.7 ± 1.9), because photon dose is density-tolerant while proton range integrates density (Fig. 5).

3.4 Diagnosing the hidden-test drop: MR intensity shift

The gap between cross-validated (91.1) and hidden-test (84.1) gamma is itself a result worth dissecting, because it is the failure mode every clinical MR-only deployment will face. Three observations triangulate it. First, synthetically shifting our validation MRs (bias/gamma/scale perturbations at magnitudes estimated from the test-set image statistics we could observe) reproduced a drop of the observed size, and attributing it stage-by-stage placed most of it in the shared tissue *classifier*, not the refiner or dose network. Second, retraining the classifier with strong intensity-shift augmentation recovered a substantial part of the board drop on the sister proton-MRI task (+3.1 board gamma) and is deployed here. Third, the recovery is partial and our synthetic shifts were harsher than the real ones, a calibration gap we report so others can budget for it: internal shift probes over-predicted the recoverable gain by roughly $4\times$. The residual hidden-test gap after augmentation is consistent with a mixture of residual intensity shift and true anatomical distribution shift in the larger test cohort.

3.5 Runtime engineering

Runtime is double-weighted, and this submission held the fastest runtime on the preliminary board at submission time (94.8 s; the compressed-output variant of

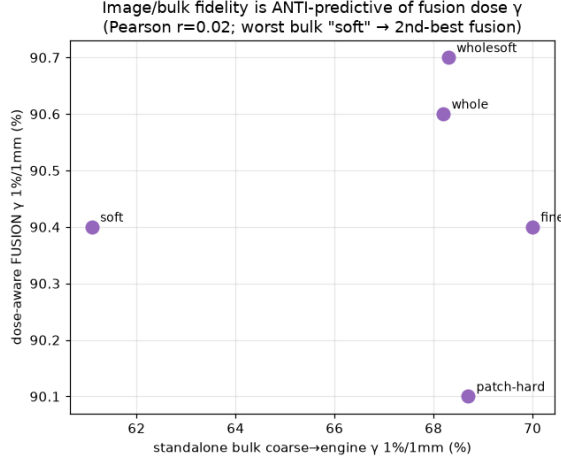


Fig. 4. Image fidelity does not predict dose accuracy. Each point is one coarse-prior variant of the synthesis chain: *soft* (soft-tissue-only bulk map), *whole* (whole-image classifier), *wholesoft* (whole-image classifier, soft-tissue bulk values), *fine* (finer tissue classes), *patch-hard* (patch-based classifier, hard labels). Horizontal axis: the variant’s *standalone* quality, the plan γ 1%/1 mm obtained when its bulk coarse CT is fed directly to the dose engine. Vertical axis: the *fusion* accuracy, the plan γ of the full dose-aware pipeline built on that variant (jointly trained synthesizer + dose network). Standalone quality spans 9 points but fusion accuracy varies by only 0.6 (Pearson $r=0.02$): the jointly trained refiner recovers dose-relevant density regardless of the coarse prior.

the same model ran 124.7 s, and margin 24 added a further ~ 30 s; both rejected by measurement). Photon-MRI adds one patient-level stage to the Photon-CT path; the synthesizer runs once per source image (~ 2 s, of which the two whole-image resamples and two network forwards are profiled separately in the container log); after which the per-CP loop is identical: aperture-bbox crops (margin 16, selected on full-grid metrics), GPU-resident channels, compiled dose net, cutoff-zeroed uncompressed output (photon jobs are compute-bound; an earlier compressed variant was 30 s slower). Two accuracy-relevant deployment details: the naive prior at deploy is the same skin-entry-gated surrogate the network was trained with (the CT container’s build-up variant is *not* interchangeable), and the whole-image classifier requires the same inference mode as training, a sliding-window variant collapsed lung on some grids in the sister proton container, a failure mode we gate for explicitly.

4 Discussion

The central lesson of this task is that *dose-aware synthesis works and image metrics mislead*: $+8.1 \gamma$ from bulk to learned sCT, $+6.2$ more from making synthesis serve the dose (Table 2), while image fidelity was non-predictive of the outcome

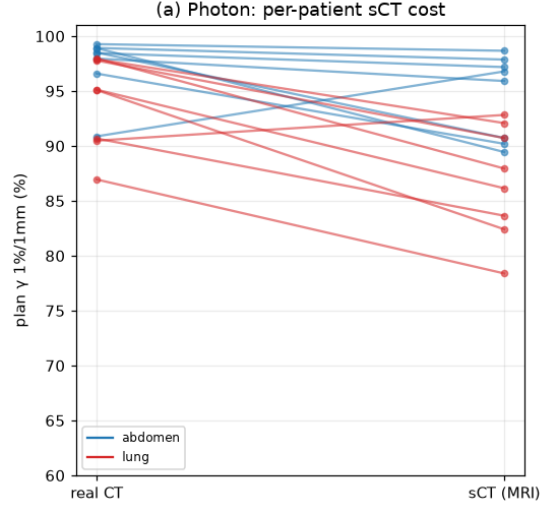


Fig. 5. Per-patient synthesis cost on this task (real CT vs. sCT input, development fold; blue abdomen, red lung): photon drops are small and *mixed*; photon dose is density-tolerant. The corresponding proton cost is $\sim 2\times$ larger (see text).

($r=0.02$). The remaining gap to the real-CT ceiling (4.7 points cross-validated) is a genuine synthesis cost, concentrated in lung at 0.35 T where the MR simply carries little density information; better segmentation and robustness augmentation both failed to close it, suggesting a data limit (field strength, cohort size) rather than a capacity limit. **Limitations and generalizability.** Single benchmark, single 0.35 T scanner; the hidden-test intensity shift we diagnosed is a preview of the cross-device shift any clinical deployment faces, and our mitigation (shift-augmented classification) recovers only part of it. The ground truth is the CT-computed dose mapped to the MR, so residual CT–MR registration error is label noise. MR-linac photon planning with bulk or sCT densities is already clinically accepted; our contribution is a stronger, dose-optimized MR-to-dose path and a caution about how to select its components.

Why an intermediate sCT at all? Three design questions deserve explicit answers. First, the intermediate density image exists because it is the *interface to the physics*: the analytical operator (radiological depth, fluence attenuation) consumes a density map, and it is this operator that supplies the residual-learning advantage and the data efficiency of the whole framework; a direct MR-to-dose network would have to relearn that physics from 75 patients. Second, the synthesizer is partially transferred: the refiner is pretrained on 166 field-matched SynthRAD2025 MR–CT pairs before dose-aware joint training. Third, the dose-aware sCT genuinely *diverges* from an imaging-optimized sCT: trained through the dose loss, the synthesizer accepts a worse image error where the dose does not need fidelity and spends its capacity where the dose does; this is precisely

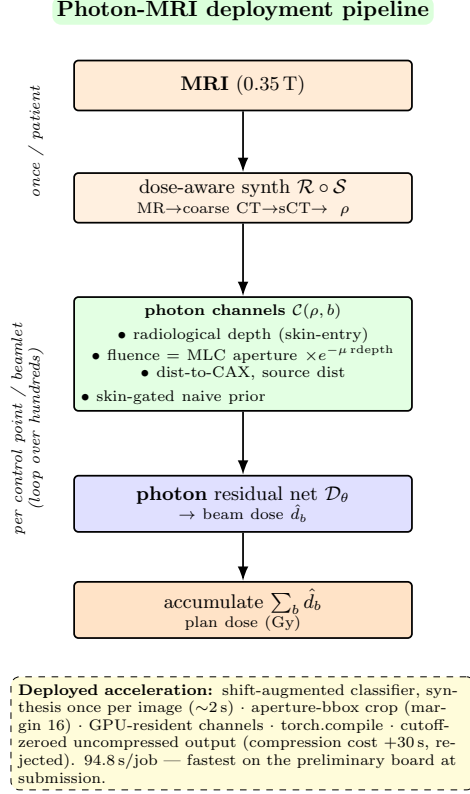
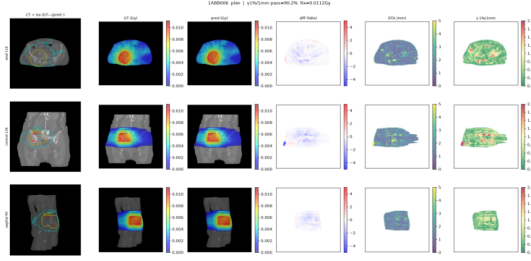


Fig. 6. The deployed Photon-MRI container: per patient, the density stage runs once; the differentiable-channels $\rightarrow$ residual-net loop runs per control point/beamlet. The callout lists the deployed acceleration measures and their measured effects.

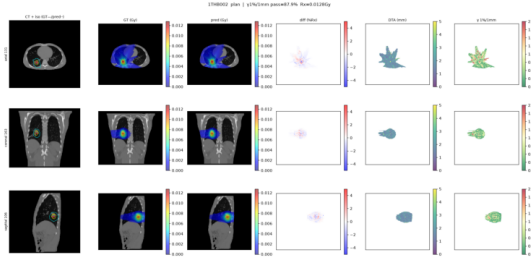
why image fidelity stops predicting dose accuracy (Sec. 3.3), and it means a dose-aware sCT should not be reused as a diagnostic-quality image. Whether a fully sCT-free direct MR-to-dose model can match this pipeline remains open (we did not run that controlled experiment) and is our first future-work item.

Future work. (i) *sCT-free direct MR-to-dose*: the fidelity result implies the sCT is only a means to an end; a network consuming the MR (plus geometry channels computed on a provisional density) end-to-end may bypass the synthesis bottleneck entirely, and the strongest competing entries in this challenge appear to point the same way. (ii) *Shift-robust synthesis by design*: replace heuristic intensity augmentation with test-time intensity-normalization or physics-based bSSFP signal modelling, targeting the classifier that our attribution identified as the shift-sensitive stage. (iii) *Speed*: our distilled base-32 photon student ($2.1\times$, teacher-exceeding on CT) transfers to this task only after finetuning on synthesis-domain channels, a domain-adapted student is trained and, if it passes our no-

Abdomen (1ABB006) — plan $\gamma 1\%/1\text{mm} = 90.2\%$



Lung (1THB002) — plan $\gamma 1\%/1\text{mm} = 87.9\%$



Hard case (1THB143) — plan $\gamma 1\%/1\text{mm} = 67.7\%$

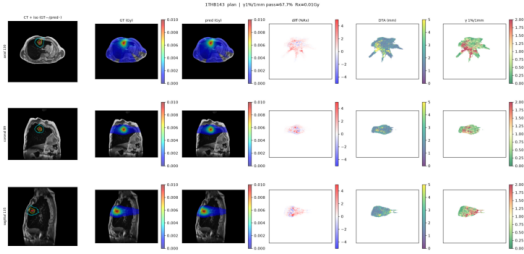


Fig. 7. Qualitative Photon-MRI results on three cases spanning the difficulty range: abdomen (1ABB006, plan $\gamma 1/1 = 90.2$), lung (1THB002, 87.9), and the geometrically hard case 1THB143 (67.7; the worst patient in all four challenge tasks simultaneously, yet largely passing at 3%/3 mm: small-magnitude errors at steep lung–tissue interfaces). Columns: image with GT (solid) and predicted (dashed) isodose, MC dose, predicted dose, difference, distance-to-agreement, local $\gamma 1\%/1\text{ mm}$.

accuracy-loss gate, halves the per-CP cost. (iv) *Higher field and larger cohorts*: the lung synthesis cost tracks the information limit of 0.35 T bSSFP; 1.5 T MR-linac data is the clean test of how much of the remaining 4.7-point ceiling gap is physics rather than data.

5 Author Contributions (CRediT)

Kai Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Meixu**

Chen: Software (deployment and runtime optimization), Investigation (acceleration study), Validation, Writing – review & editing. **Rui Yang:** Conceptualization, Methodology, Writing – review & editing.

6 Other Information

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